

HR Analytics

Workforce, Attrition & Employee Retention Analysis

Project Type: HR Analytics | SQL + Power BI

Tools: MySQL | Power BI | DAX | Excel | GitHub

Dataset: 22,214 employee records

Project Workflow

Raw HR Data → Data Cleaning → Data Validation → SQL Analysis → Advanced SQL → DAX → Power BI
Dashboard → Business Insights

1. Executive Summary

This project analyzes **22,214 employee records** to understand workforce composition, employee demographics, hiring patterns, employee tenure, and employee attrition.

The project was developed as an end-to-end analytics workflow: **Data Import** → **Data Cleaning** → **Data Validation** → **SQL Analysis** → **Advanced SQL** → **DAX** → **Power BI Dashboard** → **Business Insights**.

The analysis found an overall termination rate of approximately **17.2%**, with meaningful differences across departments, job titles, gender groups, and work locations.

2. Business Problem

HR teams need to understand not only how many employees work in an organization, but also where employees are concentrated, how demographics are distributed, how hiring has changed, how long employees remain, and which departments or roles may require additional retention analysis.

The purpose of this project was to build a data-driven HR analysis that answers these questions using SQL and Power BI.

3. Project Objectives

- Determine the total workforce size.
- Analyze employee demographics.
- Understand department-wise workforce distribution.
- Analyze employee age distribution.
- Identify hiring patterns over time.
- Calculate employee tenure.
- Measure employee termination rates.
- Compare attrition across departments, gender groups, and work locations.
- Identify the most common job titles and roles with comparatively high termination rates.
- Build an interactive Power BI dashboard.
- Convert analytical findings into actionable HR insights.

4. Dataset

The dataset contains **22,214 employee records**.

Field	Description
emp_id	Unique employee identifier
first_name / last_name	Employee name fields
birthdate	Date of birth
gender	Employee gender
race	Race/ethnicity
department	Employee department
jobtitle	Employee job title
location	Headquarters or Remote

Field	Description
hire_date	Employee hiring date
termdate	Employee termination date
location_city / location_state	Employee location
age	Calculated employee age

5. Data Preparation & Cleaning

Database Creation

CREATE DATABASE hranalysis;
USE hranalysis;

Employee ID Correction

ALTER TABLE hr
CHANGE COLUMN id emp_id VARCHAR(20) NULL;

Date Standardization

Birthdates and hire dates were standardized with STR_TO_DATE() and converted to the MySQL DATE data type.

Termination Date Cleaning

Timestamp-style termination values were converted to dates, while blank termination values were converted to NULL so that termdate IS NULL could represent active employees.

Age Calculation

ALTER TABLE hr ADD COLUMN age INT;
UPDATE hr
SET age = TIMESTAMPDIF(YEAR, birthdate, CURDATE());

6. Data Validation

Duplicate records were checked and no duplicates were identified.

Validation Check	Result
Missing employee IDs	0
Missing birthdates	0
Missing hire dates	0
Missing departments	0
Termination date earlier than hire date	0

7. SQL Analysis

The SQL analysis progressed from basic descriptive questions to moderate and advanced HR analytics.

Total Employees

SELECT COUNT(emp_id) AS total_employees
FROM hr;

Result: **22,214 employees.**

Gender Distribution

Gender	Employees
Male	9,328
Female	8,455

Gender	Employees
Non-Conforming	502

Race/Ethnicity Distribution

Race/Ethnicity	Employees
White	6,328
Two or More Races	3,648
Black or African American	3,619
Asian	3,562
Hispanic or Latino	2,501
American Indian or Alaska Native	1,327
Native Hawaiian or Other Pacific Islander	1,229

Age Distribution

Employee ages in the analyzed dataset ranged from approximately **23 to 60 years**, providing a broad view of the workforce across career stages.

8. Department Analysis

Department	Employees
Engineering	6,686
Accounting	3,333
Sales	1,832
Human Resources	1,807
Training	1,692
Services	1,686
Business Development	1,642
Research & Development	1,084
Support	954
Product Management	641
Marketing	494
Legal	311
Auditing	52

Key finding: Engineering was the largest department with **6,686 employees**. Auditing was the smallest with **52 employees**.

9. Employee Attrition Analysis

The overall termination rate calculated during the analysis was approximately **17.2%**.

Department	Employees	Terminated	Rate
Auditing	52	12	23.08%
Legal	311	63	20.26%
Research & Development	1,084	212	19.56%

Department	Employees	Terminated	Rate
Support	954	182	19.08%
Sales	1,832	335	18.29%
Product Management	641	114	17.78%
Engineering	6,686	1,185	17.72%
Accounting	3,333	586	17.58%
Services	1,686	293	17.38%
Training	1,692	291	17.20%
Human Resources	1,807	309	17.10%
Business Development	1,642	275	16.75%
Marketing	494	72	14.57%

10. Key Attrition Findings

Auditing — Highest Termination Rate

Auditing had the highest termination rate at **23.08%**, with 52 employees and 12 terminations. Because the department is small, the percentage should be interpreted cautiously.

Legal — Potential Retention Concern

Legal recorded a **20.26%** termination rate and **14.09 years** average tenure, making it an important department for further retention investigation.

Engineering — Largest Workforce Impact

Engineering recorded **1,185 terminations**, the highest absolute number in the organization, but its termination rate was **17.72%**. This demonstrates why both counts and rates matter.

Marketing — Lowest Termination Rate

Marketing had the lowest department termination rate at **14.57%**, with 494 employees and 72 terminations.

11. Gender-Based Attrition

Gender	Employees	Terminated	Rate
Female	10,321	1,866	18.08%
Male	11,288	1,960	17.36%
Non-Conforming	605	103	17.02%

Female employees recorded the highest observed termination rate at **18.08%**. The differences are relatively modest and should not be interpreted as causal without further analysis.

12. Location-Based Attrition

Location	Employees	Terminated	Rate
Headquarters	16,715	3,005	17.98%
Remote	5,499	924	16.80%

Headquarters employees had a termination rate approximately **1.18 percentage points higher** than Remote employees.

13. Employee Tenure Analysis

Average employee tenure was calculated using hire and termination dates. For active employees, the current date was used as the end date. The overall average tenure was approximately **14.80 years**.

Department	Average Tenure
Auditing	15.23 years
Product Management	15.01 years
Engineering	14.85 years
Business Development	14.84 years
Training	14.84 years
Research & Development	14.82 years
Accounting	14.79 years
Support	14.77 years
Human Resources	14.76 years
Sales	14.74 years
Services	14.73 years
Marketing	14.48 years
Legal	14.09 years

14. Hiring Trend Analysis

Hiring activity was analyzed across **2000–2020**. The lowest observed hiring count was **220 employees in 2000**, while the highest was **1,147 employees in 2018**. Most years after 2001 showed relatively stable hiring levels.

15. Top Job Titles

Rank	Job Title	Employees
1	Research Assistant II	754
2	Business Analyst	708
3	Human Resources Analyst II	613
4	Research Assistant I	538
5	Account Executive	505
6	Data Visualization Specialist	457
7	Staff Accountant I	441
8	Human Resources Analyst	408
9	Software Engineer I	397
10	Systems Administrator I	374

16. High-Termination Job Titles

Job Title	Employees	Terminated	Rate
Financial Analyst	68	17	25.00%
Systems Administrator IV	88	22	25.00%
Customer Success Manager	214	53	24.77%

Job Title	Employees	Terminated	Rate
Accounting Assistant IV	95	23	24.21%
Developer III	83	20	24.10%
Junior Trainer	181	43	23.76%
Paralegal	64	15	23.44%
Internal Auditor	52	12	23.08%
Web Developer III	71	16	22.54%
Senior Quality Engineer	152	34	22.37%

17. Advanced SQL Analysis

Advanced SQL techniques were used to move beyond simple aggregation.

- CTEs were used to structure multi-step analysis and improve readability.
- Window functions such as RANK() and ROW_NUMBER() were used for ranking.
- PARTITION BY was used to perform calculations within department or other business groups.
- Department-level summaries combined employee counts, terminations, termination rates, and average tenure.

18. Power BI Dashboard

The cleaned dataset was imported into Power BI to create an interactive HR analytics dashboard.

- **Workforce Overview:** Total Employees, Active Employees, Terminated Employees, Termination Rate, Average Tenure.
- **Demographics:** Gender, race/ethnicity, age, and department analysis.
- **Attrition & Retention:** Termination rate by department, gender, and location; average tenure by department.
- **Workforce Structure:** Active employees by department, top job titles, and hiring trend by year.

19. DAX Analysis

Power BI DAX was used to calculate employee tenure dynamically using hire date, termination date, and the current date for active employees. The resulting tenure measure can respond to dashboard filters such as department, gender, location, and job title.

20. Data Quality & Future-Dated Records

The source dataset contained termination dates extending beyond the intended historical analysis period. Rather than artificially changing the underlying source data, the termination trend visualization was restricted to **31 December 2025** so that future-dated records would not distort the historical trend.

This was treated as a data-quality and visualization consideration during dashboard development.

21. Business Insights

- **Workforce:** 22,214 employees, with Engineering as the largest department at 6,686 employees.
- **Attrition:** Overall termination rate was approximately 17.2%.
- **Highest department attrition:** Auditing at 23.08%, with a small workforce of 52 employees.

- **Retention concern:** Legal combined a 20.26% termination rate with the lowest department-level average tenure of 14.09 years.
- **Workforce impact:** Engineering had the highest number of terminations at 1,185.
- **Lowest department attrition:** Marketing at 14.57%.
- **Gender:** Female employees had the highest observed termination rate at 18.08%.
- **Location:** Headquarters had a 17.98% termination rate versus 16.80% for Remote employees.
- **Tenure:** Overall average employee tenure was approximately 14.80 years.
- **Job roles:** Research Assistant II was the largest job title with 754 employees, while several roles showed termination rates above 20%.

22. Business Recommendations

- Investigate high-turnover departments such as Legal and Auditing to understand the underlying causes.
- Review high-turnover job roles for compensation, workload, career progression, management, and employee experience factors.
- Monitor Engineering closely because of its high absolute number of employee exits.
- Evaluate termination rate, termination count, and tenure together rather than relying on one KPI.
- Continue monitoring workforce and attrition trends through the Power BI dashboard.

23. Project Outcome

The project transformed raw employee data into a structured HR analytics solution: **Raw HR Data → MySQL Data Cleaning → Data Validation → SQL Workforce Analysis → Advanced SQL → DAX → Power BI Dashboard → Business Insights.**

The project demonstrates practical experience in turning raw data into business-focused analysis rather than simply writing SQL queries or creating charts.

24. Skills Demonstrated

SQL: Database creation, data cleaning, data type conversion, date manipulation, NULL handling, validation, aggregation, CASE, filtering, GROUP BY, HAVING, ordering, date functions, CTEs, window functions, RANK(), ROW_NUMBER(), and PARTITION BY.

Power BI: Data import, data modeling, KPI cards, bar charts, line charts, slicers, interactive filtering, DAX calculated columns, DAX measures, workforce dashboards, and attrition dashboards.

Business Analytics: Workforce analysis, segmentation, attrition, retention, tenure, hiring trends, KPI development, business recommendations, and data-quality assessment.

25. Conclusion

The HR Analytics project demonstrates how SQL and Power BI can be combined to answer practical HR business questions. The analysis shows that employee turnover varies across departments and job roles, while workforce size significantly affects absolute termination numbers.

The most important analytical lesson is that HR decisions should not rely on a single metric. Combining **employee count, termination count, termination rate, and average tenure** provides a more complete picture of workforce stability.

The final Power BI dashboard converts these findings into an interactive reporting tool that can support workforce monitoring and further HR analysis.

Final Project Statement

This project demonstrates an end-to-end data analytics workflow, from cleaning and validating raw employee data in MySQL to performing advanced SQL analysis and building an interactive Power BI dashboard that translates workforce and attrition data into actionable business insights.